Reduced-Order Modeling of Upper-Level Winds for Fuel-Efficient Flight Routing

POD-DMD and POD-OpInf comparison for upper-level wind prediction

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THE PROJECT

ROM Workflow

1

Step one

Collect ERA5 Wind Data

2

Step two

Construct POD Reduced Basis

3

Step three

Develop POD-DMD and POD-OpInf Models and Compare

4

Step four

Apply ROMs to Flight Routing

Primary objective

Reduced-order modeling of upper-level winds.

Secondary objective

ROM-based flight route optimization.

THE DATA

ERA5 Dataset at a Glance

3,720

Hourly wind snapshots

8,833

Spatial grid points

17,666

State variables

250 hPa

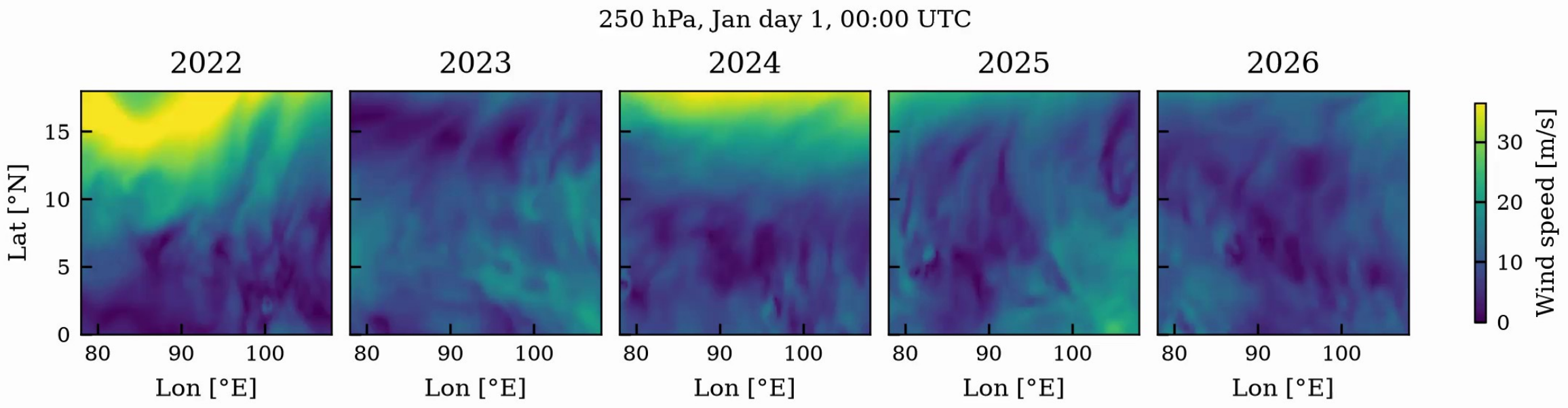
Upper-level wind data

Key details

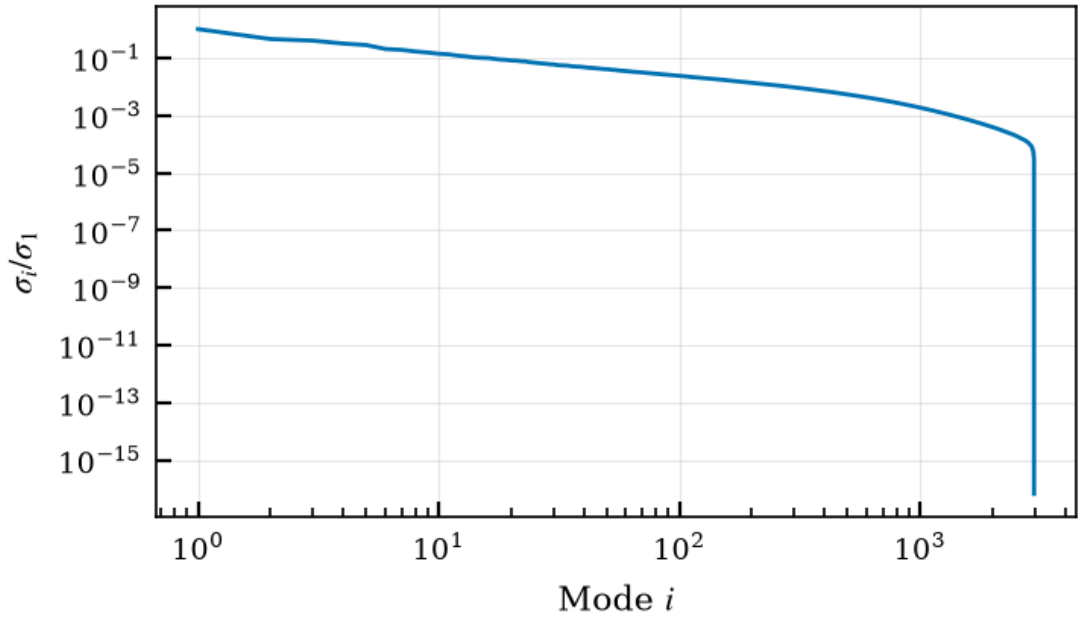
- Source: ERA5 hourly reanalysis (2022–2026).
- Pressure level: 250 hPa (~10.4 km).
- Domain: 0°–18°N, 78°–108°E.
- Resolution: 0.25° × 0.25° (8,833 grid points).
- Dataset size: 3,720 snapshots, 17,666 state variables.
- Data split: 2,976 training (2022–2025), 744 testing (January 2026).

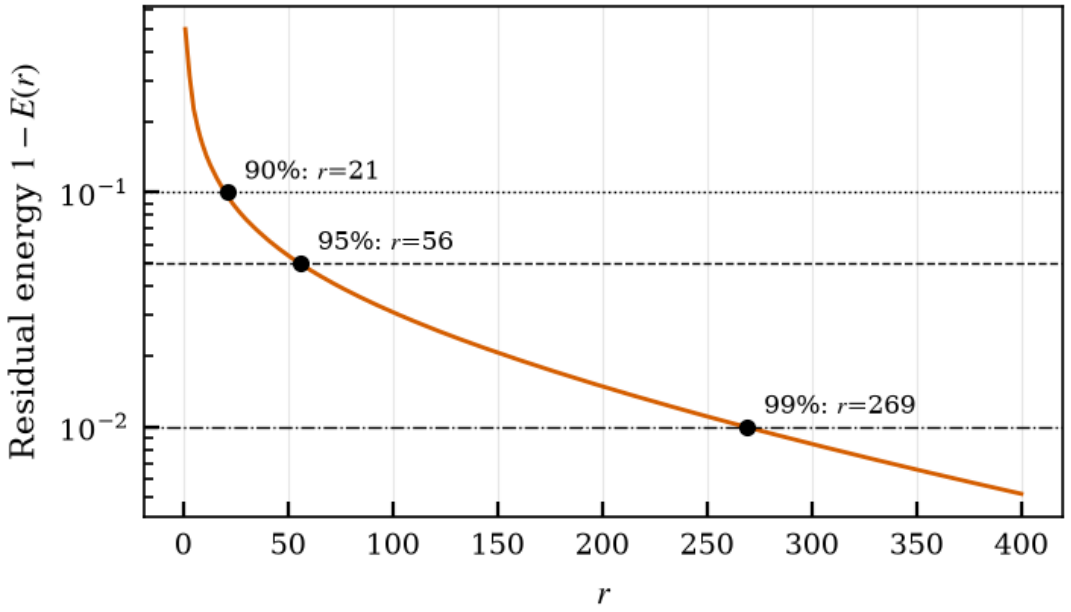
THE DATA

ERA5 Dataset at a Glance



Performing POD on training dataset

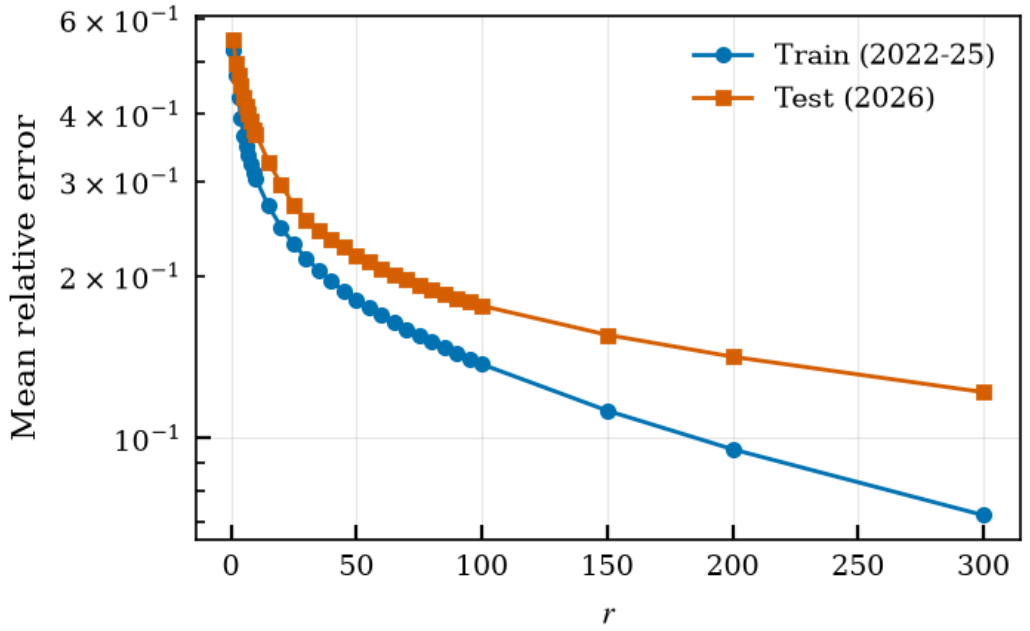


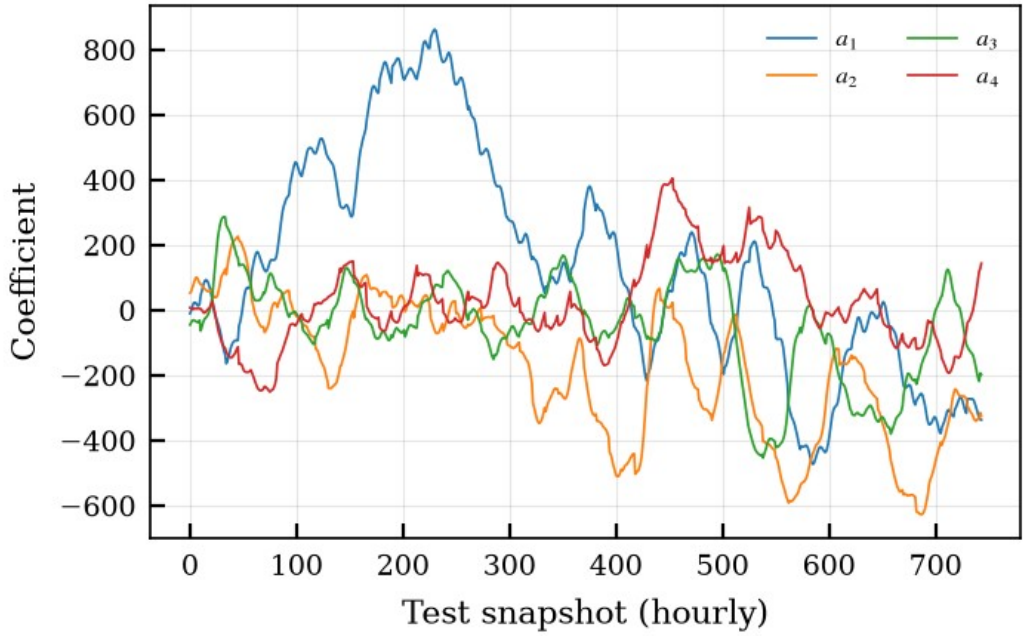


Energy(%)	90	95	99
Modes	21	56	269

The training dataset:
2022 - 2025

Testing the POD basis

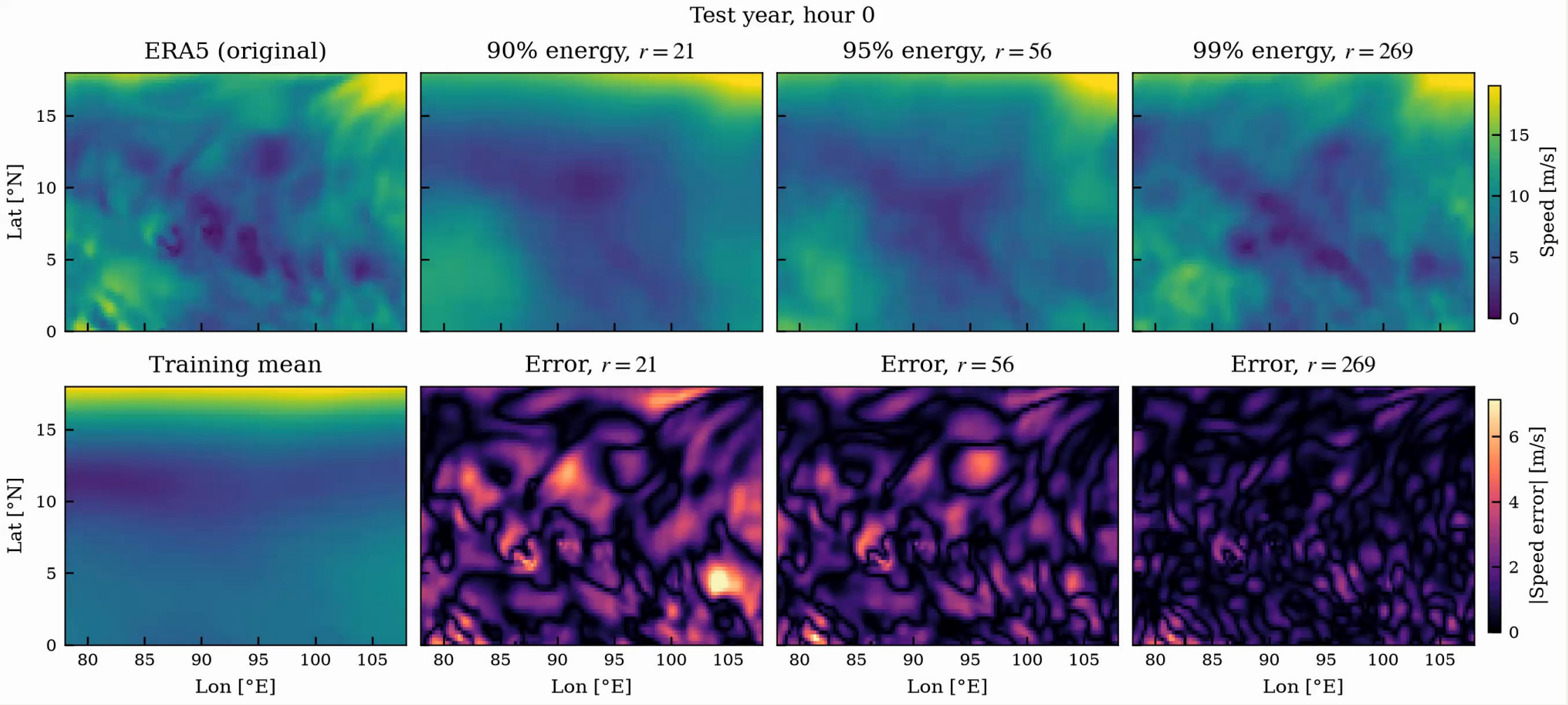




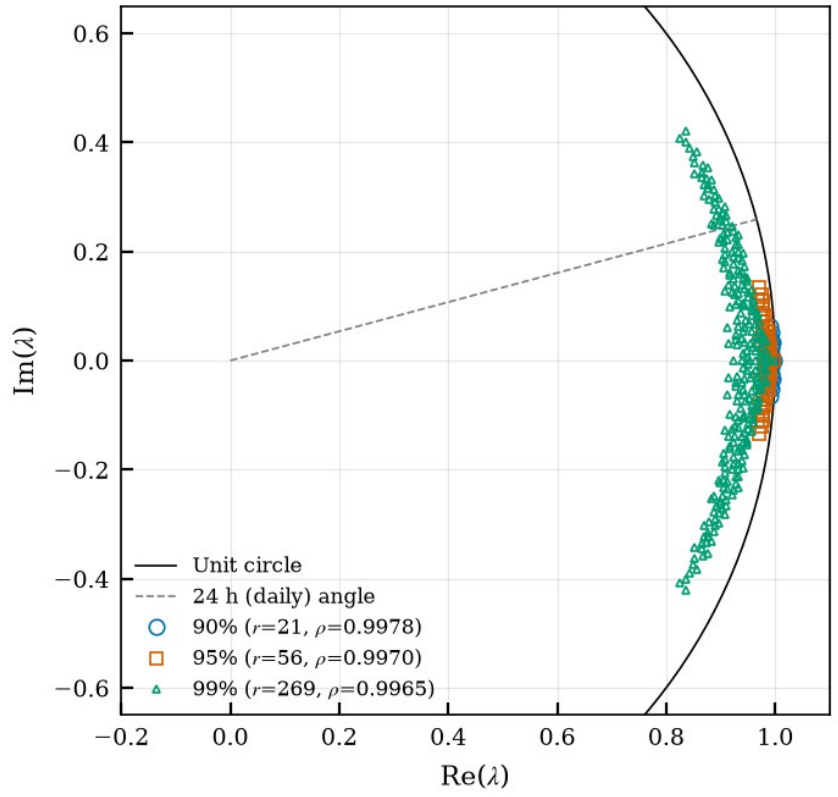
Energy(%)	90	95	99
Modes	21	56	269

The testing dataset:
2026

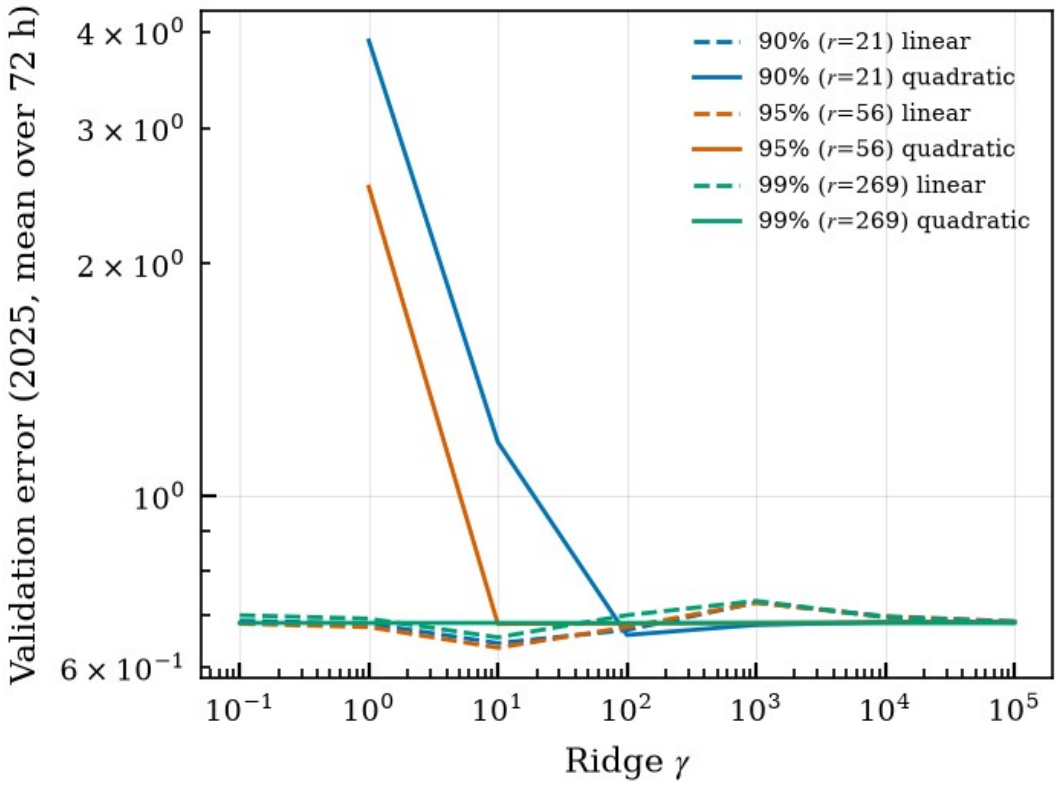
Reconstructed Test Data with POD Basis



DMD Stability and OpInf Validation



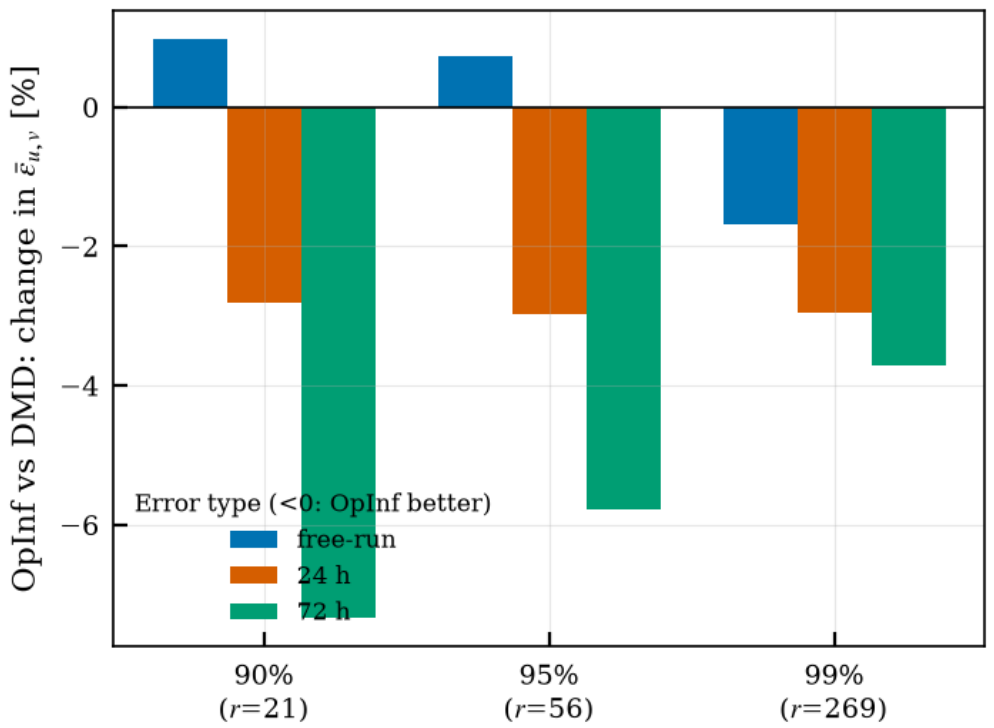
DMD is Stable



Selected value of ridge = 10

COMPARISON

Mean Absolute Error



<0

OpInf Better

7 / 9

POD-OpInf won

9

runs

RESULTS

The Key Values

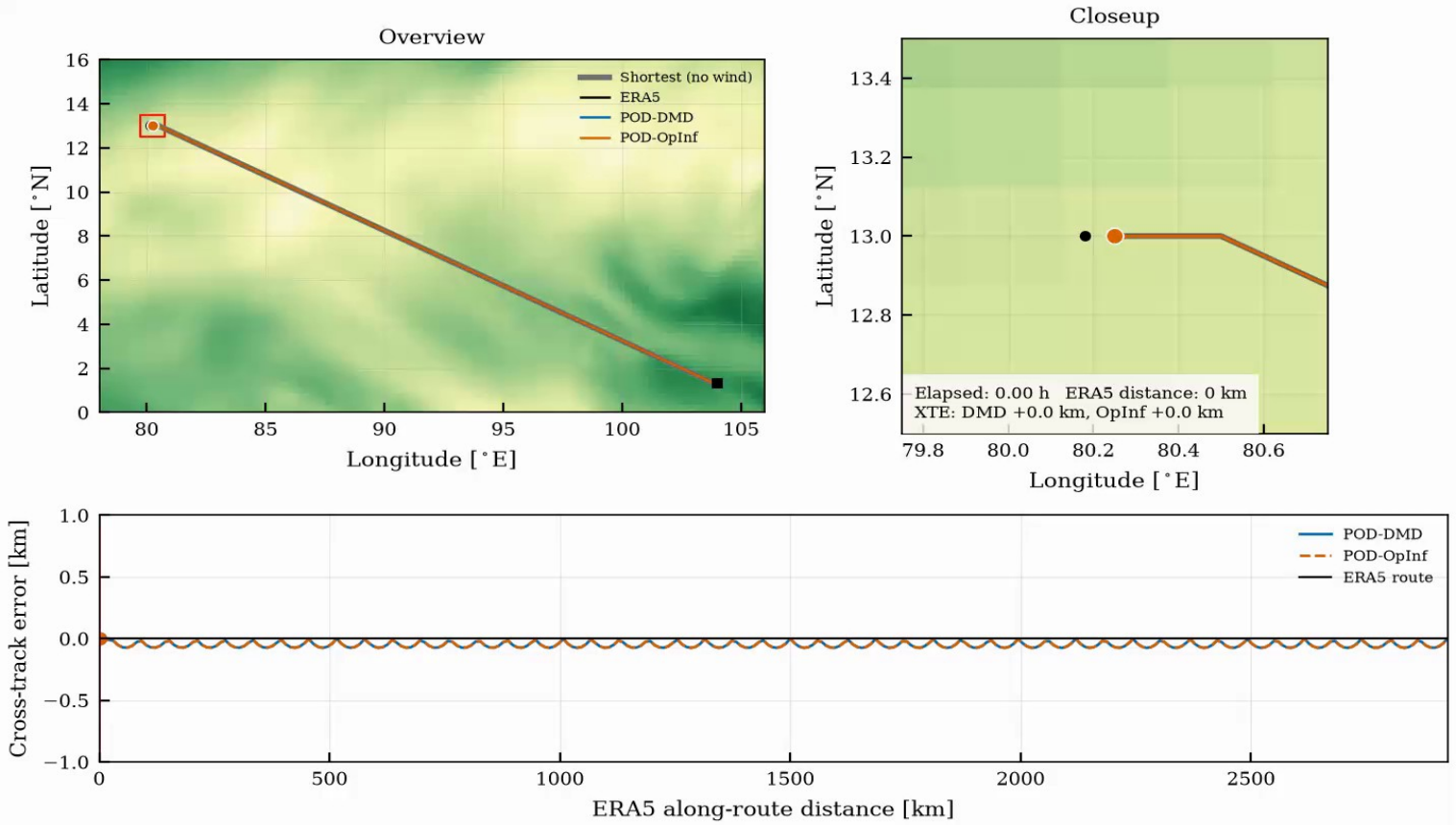
Energy	Metric	DMD	OpInf
90%	Spectral radius	0.9978	0.9944
90%	Fit time (s)	0.0056	0.0014
90%	Prediction time (s)	0.0766	0.0773
90%	Storage (MiB)	2.97	2.97
95%	Spectral radius	0.9970	0.9937
95%	Fit time (s)	0.0119	0.0037
95%	Prediction time (s)	0.0704	0.0834
95%	Storage (MiB)	7.71	7.71
99%	Spectral radius	0.9965	0.9931
99%	Fit time (s)	0.1262	0.0139
99%	Prediction time (s)	0.1215	0.1133
99%	Storage (MiB)	36.94	36.94

Observations

- OpInf takes less time to fit which means that the system comparitively more non-linear in nature.
- Spectral radius of OpInf is comparatively lesser which means slightly faster decay towards training mean.

THE APPLICATION

Flight Routing

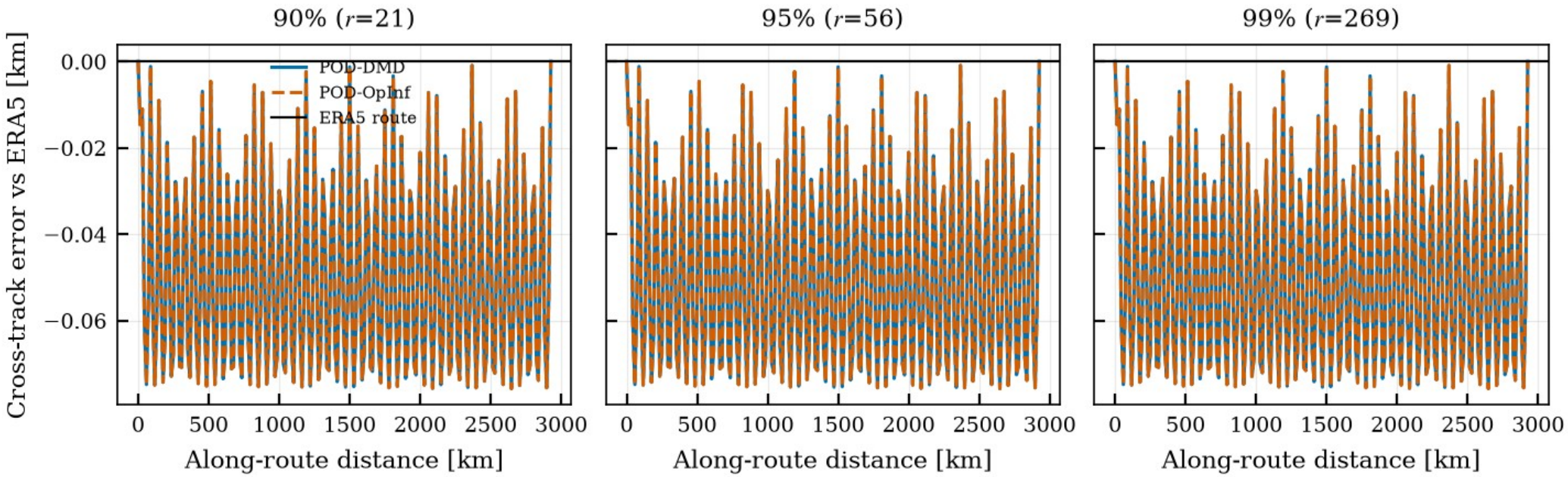


Chennai
(13.00, 80.18)

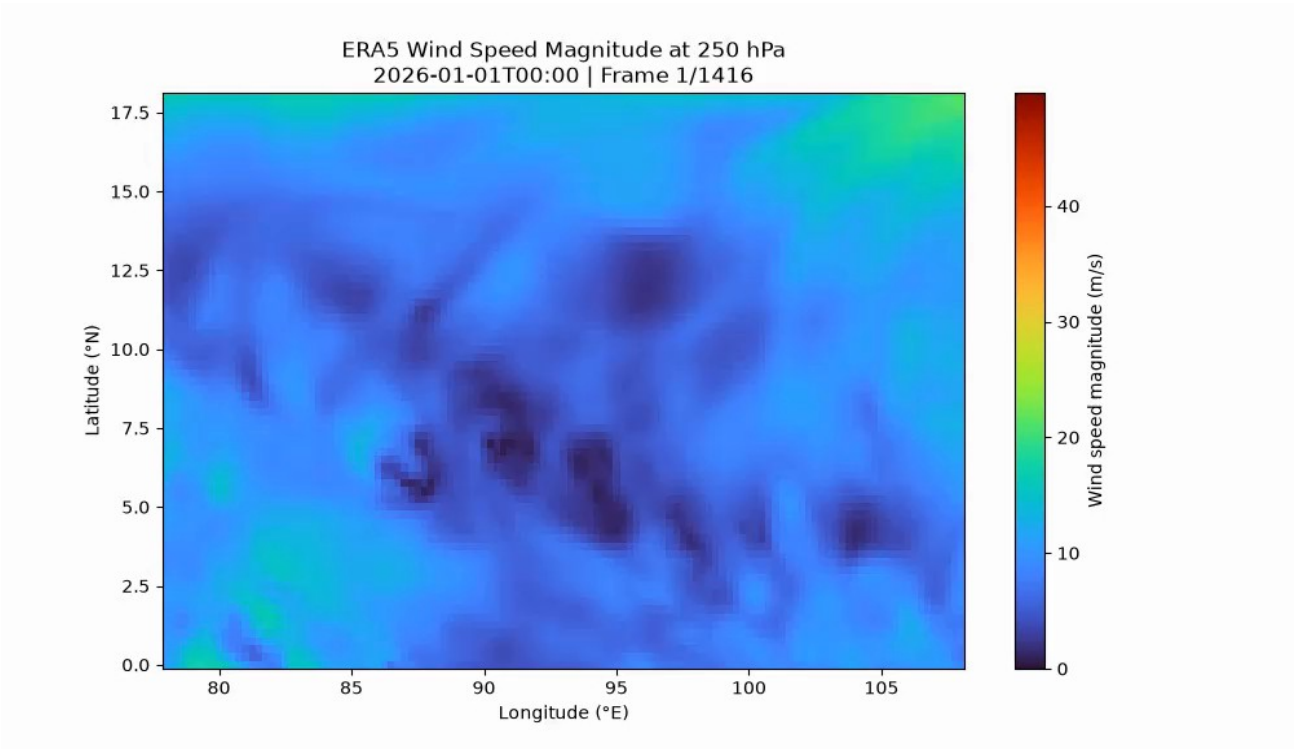
Singapore
(1.36, 103.99)

RESULTS

Cross-Track Error (Jan 13, 2026)



Thank you


ERA5 Wind Speed Magnitude at 250 hPa
2026-01-01T00:00 | Frame 1/1416
Latitude (°N)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
80
85
90
95
100
105
Longitude (°E)
Wind speed magnitude (m/s)
40
30
20
10
0